# The Theoretical Minimum

You can get a deep understanding of physics by working through a small set of canonical textbooks.

## The Undergraduate Minimum

The following books are well-written and beloved by students. They are particularly good for self-study because they are self-contained, anticipate possible misconceptions, and most importantly, contain lots of good problems. If you work through them, doing many of the problems on your own, then you'll know the core theoretical content of an undergraduate physics degree.

1. Any introductory calculus-based physics book, as listed here. They're all about equally good, so just get any one that's reasonably common.

2. *Classical Mechanics* by Taylor. A very smooth and easy introduction to intermediate mechanics, with thorough explanations and lots of problems of varying difficulty. Can be started with only ordinary calculus background, with the required vector calculus gradually introduced.

3. *Thermal Physics* by Schroeder. An extremely clean, focused, and clear introduction. Requires exposure to multivariable calculus, e.g. the multivariable chain rule and multiple integrals.

4. *Introduction to Electrodynamics* by Griffiths. The best book for undergraduate electromagnetism by a huge margin. Achieves the almost impossible combination of being required for just about every course on the subject, yet still beloved by its readers. Starts with an excellent crash course on vector calculus. Contains a rich collection of problems, and extensive, student-tested explanations. The 4th and 5th editions are equivalent.

5. *Introduction to Quantum Mechanics* by Griffiths and Schroeter. The most accessible introduction to quantum mechanics, assuming only ordinary calculus and introducing a little linear algebra when needed. Built around a core of intuition-building, elementary exercises, plus a rich variety of harder problems. The 3rd edition is significantly better than the 2nd.

The four books I recommend above are *exactly* the same as those recommended in Susan Rigetti's excellent guide, and by AI if you ask it neutrally (as of mid-2026). This isn't a coincidence: there really is widespread consensus that these are great books to start from![1]

There are of course more "hardcore" books, such as Landau and Lifshitz for mechanics and Shankar for quantum mechanics. These are also great, but not as suitable for self-study because they omit context a beginner needs, lack rich practice problems, and take mathematical background for granted. Like any book, reading them will enrich your understanding, but I recommend mastering the books above first. After you do, these harder books will be easily approachable.

## Electives

Here are some excellent, gentle first books in several subfields of physics. None of them will get you all the way, but once you finish one, you'll have a clearer idea of what to do next. All of them should be readable after completing the undergraduate minimum.

- Astrophysics: *An Introduction to Modern Astrophysics* by Carroll and Ostlie.

[1] There's also a series of four new books from David Tong, which seem promising, though I haven't read them.

- Biophysics: *Biological Physics: Energy, Information, Life* by Nelson.
- Cosmology: *Introduction to Cosmology* by Ryden.
- Condensed matter: *The Oxford Solid State Basics* by Simon.
- Continuum mechanics: *Physics of Continuous Matter* by Lautrup.
- Chaos theory: *Nonlinear Dynamics and Chaos* by Strogatz.
- Particle physics: *Introduction to Elementary Particles* by Griffiths (for the basics of computations) or David Tong's Particle Physics lecture notes (for a qualitative modern overview).
- Plasma physics: *Introduction to Plasma Physics* by Chen.
- Quantum info: *Quantum Computation and Quantum Information* by Nielsen and Chuang.
- Quantum optics: *Introductory Quantum Optics* by Gerry and Knight.
- String theory: *A First Course in String Theory* by Zwiebach.

## The Graduate Minimum

In America, a PhD in physics typically begins with seeing all the undergraduate material again at a higher level. Here there's a standard set of recommendations that hasn't changed for 40 years. These classics are solid, but I think some have been improved upon by more modern sources, which are more relevant to current research and easier to understand. My personal recommendations are:

6. *Modern Classical Mechanics* by Helliwell and Sahakian. (Improved version of Goldstein.)

   The last third of Goldstein covers the canonical formalism in depth. This is crucial for some of the most beautiful applications of mechanics, but isn't used in many fields at all. Helliwell and Sahakian's new book contains many fresh examples and problems, with relevance to modern research, while also emphasizing the aspects of classical mechanics that are relevant to quantum mechanics.

7. *Modern Electrodynamics* by Zangwill. (Improved version of Jackson.)

   Jackson's book is excellent and thorough, and has been the standard since the 1960s, but its topics are somewhat out of date. The first third covers boundary value problems in great detail, but most physicists today would just use a computer instead. The last third covers topics which are most useful for building particle accelerators, reflecting Jackson's research. Zangwill's new book has more discussion of how equations are derived and what they mean, and includes glimpses into how electromagnetism is used across all of physics. The only downside is that Zangwill defers relativity and Lagrangians until the end of the book. This makes it less useful as a stepping stone towards quantum field theory.

8. *Statistical Physics* (two volumes) by Kardar. (Improved version of Pathria/Huang/Reif.)

9. Quantum Mechanics lecture notes by Littlejohn. (Improved version of Sakurai.)

   Sakurai tragically passed away before completing a draft of his book, so the rest was filled in by assistants using his handwritten notes. This led to dozens of "bugs" throughout, like nonsensical problems, undefined or ambiguous notation, and dubious explanations, many of which remain in the 3rd edition today. (It's given me a lot of headaches, both when learning the subject and working as a TA.) Despite these flaws, it was still the most complete and modern option in the 1980s, and remains the traditional standard in American graduate courses. Littlejohn's notes cover similar material, but have been refined over decades of teaching graduate quantum mechanics, and every aspect of them is perfectly clear.

### Further Specializations

Students often ask for a list of books to read to be totally "prepared" for research, but this is misguided: even within a subfield (such as particle physics), the answer depends entirely on the research problem you end up working on. There's too much in each subfield for any one person to learn. The right path is to master the fundamentals above, then get a research mentor who will direct you to an interesting problem, and help you figure out what you need to learn to attack it.

Alternatively, if you don't intend to do research in physics, and only want to study it for fun, then the right path forward is to figure out precisely why you're going to the trouble of doing so. Are you fascinated by the story of physics as a science? If so, you're well-prepared to read about the history of physics, either through the works of historians or directly from the primary sources. Do you enjoy the struggle of cracking tough problems? You can find many more such problems throughout the literature, or you might want to start formulating and solving your own. Do you just want to know what's going on at the research frontier? The next time you see a mindblowing but vague news article about the latest breakthrough, you'll be prepared to go right to the source and see for yourself. The "minimum" knowledge above will not be enough to understand those papers immediately, but it will be enough for you to figure out how to learn the parts you're missing in a reasonable amount of time. The foundation is set.